“Seminars in Biomedical Engineering”

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DETECTING CONSCIOUSNESS IN THE INTENSIVE CARE UNIT WITH TMS-EEG: STATE OF THE ART AND LIMITATIONS

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Abstract

Bedside behavioral examination is the gold standard for clinical assessment of consciousness in patients with severe brain injury. However, deficits in sensory, executive, and motor functions may limit the patient’s behavioral responses, thus confounding the assessment. As such, reliable biomarkers are needed to guide clinicians caring for and attempting to prognosticate for brain-injured patients with DoC. Transcranial magnetic stimulation combined with electroencephalography (TMS-EEG) is a neurophysiological tool able to record the EEG response of the thalamocortical system to a direct and non-invasive stimulation of the cerebral cortex, bypassing sensory pathways and subcortical structures, and thus overcoming these limitations. The Perturbational Complexity Index (PCI), a measure based on EEG responses to TMS, has been recently suggested as a reliable marker to detect consciousness in patients who have recovered to a minimally conscious state (MCS) with an unprecedented sensitivity of the 94%. Despite promising, several barriers prevent the use of the TMS-EEG for the evaluation of consciousness as a

routine in the clinical setting, such as the long duration of the assessments and the lack of standardization. Here we present the state of the art, the advantages and the limitations of this technique, and the future steps that need to be addressed to become clinically applicable.

Speaker's Bio

Dr. Matteo Fecchio received his bachelor's degree in Biomedical Engineering from the University of Pavia, Italy, and his Ph.D. degree in Physiology from the University of Milan, Italy. He is currently an Instructor in Neurology at Massachusetts General Hospital. During his Ph.D. and post-doctoral fellowship at the laboratory of Prof. Marcello Massimini, he investigated the role of bistable cortical dynamics in the reduction of the level of consciousness occurring in unresponsive wakefulness syndrome and slow-wave sleep as opposed to wakefulness. To this aim, he performed measurements coupling transcranial magnetic stimulation with electroencephalography (TMS-EEG) in healthy controls and patients with a disorder of consciousness (DoC). He developed several tools to analyze EEG responses to TMS. Dr. Fecchio's current research activity at the Laboratory for NeuroImaging of Coma and Consciousness, directed by Dr. Brian L. Edlow, focuses on the application of brain-based measures for detecting signs and correlates of consciousness based on task-based EEG and TMS-EEG. Dr. Fecchio co-led an international multi-site initiative to validate the Perturbational Complexity Index (PCI), a measure of brain complexity derived from TMS evoked potentials. He is currently training post-doctoral fellows and research technicians to perform TMS-EEG measurements in patients with DoC at MGH with the long-term goal of creating a clinical program to assess patients' levels of consciousness through PCI.